

Revised Generation Interconnection System Impact Study Report

For Queue Project Q14-0420

Cement Plant 138 kV
January 19, 2017

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q14-0420. The Interconnection Customer proposes a 164.9 MW solar photovoltaic facility interconnecting at the Cement Plant 138 kV substation in the Virginia area. The request was received on December 28, 2014. The study evaluated system intact and single contingency conditions using the regional planning case, including 172 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 19 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$9,824,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Briar Substation-Fraser 138 KV line	Hinton Ap-Manatee 345 KV	126.6	201	12.4
Lena-Ou Spirit Wind Farm 138 KV	New Hope Ap-Barr Lake Substation 345 KV Line	117.1	219	10.6
George Dickie-Cement Plant 138 KV	Atoka-John Sevier 345 KV	116.0	233	34.6
Lena Substation-Optical Coating Lab 138 KV	Barr Lake-New Hope Ap 345 KV	115.9	224	10.6
Tom Watkins-Kimper 138 KV	Catawba-Atoka 345 KV	114.6	232	15.4
Nespelem-Optical Coating Lab 138 KV	Barr Lake-New Hope Ap 345 KV L	112.4	244	9.1
Halls Ridge-Pringle Substation 138 KV	New Hope Ap-Manate 345 KV line	111.7	189	6.1
Friendship-Vacherie 138 KV	Kean Mtn And Grassy Crk-Barr Lake 345 KV line	111.4	213	10.6
Nespelem Substation-Piute Substation 138 KV line	Barr Lake-New Hope Ap 345 KV	110.8	251	9.1
Rhineland Paper Substation-Taloga 138 KV	Barr Lake-New Hope Ap 345 KV Line	108.9	218	5.8
George Dickie-Southcenter 138 KV	Atoka-John Sevier 345 KV Line	108.7	244	34.2
Bandhead-S Princeton 138 KV line	Manatee-New Hope Ap 345 KV	105.4	243	6.1
Bandhead-New Hope Ap 138 KV	New Hope Ap-Manatee 345 KV	105.1	259	6.1
Pitchgen Substation-Briar 138 KV Line	Manatee-Hinton Ap 345 KV Line	105.1	240	12.4

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
South Princeton Substation-Halls Ridge 138 KV L	New Hope Ap-Manatee 345 KV	104.3	224	6.1
Coyote-Permc 138 KV	Kean Mtn And Grassy Crk-Ioco 138 KV line	103.5	334	5.1
Permac-Coal Creek 138 KV	Kean Mn And Grassy Creek-Ioco 138 KV	103.1	342	5.1
Sysquc-Rochester 138 KV	Lowmoor-Hinton Ap 345 KV line	102.2	177	5.2
Clinch Riv-Carlton Hills 345 KV	Kean Mtn And Grassy Creek 138/1 kV Transformer	102.1	984	19.3

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Briar Substation-Fraser 138 KV line	Rebuild line with higher-capacity conductor	\$4,348,000	8% (\$344,000)
Lena-Ou Spirit Wind Farm 138 KV	Rebuild line with higher-capacity conductor	\$7,075,000	8% (\$573,000)
George Dickie-Cement Plant 138 KV	Rebuild line with higher-capacity conductor	\$7,417,000	34% (\$2,543,000)
Lena Substation-Optical Coating Lab 138 KV	Rebuild line with higher-capacity conductor	\$11,776,000	8% (\$954,000)
Tom Watkins-Kimper 138 KV	Rebuild line with higher-capacity conductor	\$7,539,000	6% (\$444,000)
Optical Coating Lab-Nespelem 138 KV Line	Rebuild line with higher-capacity conductor	\$5,902,000	8% (\$498,000)
Halls Ridge-Pringle Substation 138 KV	Rebuild line with higher-capacity conductor	\$9,437,000	7% (\$669,000)
Friendship-Vacherie 138 KV	Rebuild line with higher-capacity conductor	\$7,163,000	11% (\$817,000)
Piute-Nespelem 138 KV	Rebuild line with higher-capacity conductor	\$9,950,000	8% (\$839,000)
Rhineland Paper Substation-Taloga 138 KV	Reconductor line	\$6,037,000	7% (\$423,000)
George Dickie-Southcenter 138 KV	Reconductor line	\$1,416,000	31% (\$440,000)
Bandhead-S Princeton 138 KV	Reconductor line	\$4,141,000	7% (\$293,000)
Bandhead-New Hope Ap 138 KV	Reconductor line	\$641,000	7% (\$45,000)
Briar-Pitchgen 138 KV	Reconductor line	\$1,574,000	8% (\$125,000)
South Princeton Substation-Halls Ridge 138 KV L	Reconductor line	\$2,625,000	7% (\$186,000)
Coyote-Permac 138 KV line	Reconductor line	\$7,829,000	2% (\$176,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Permac-Coal Crk 138 KV	Reconductor line	\$5,359,000	2% (\$120,000)
Sysqac-Rochester 138 KV	Reconductor line	\$1,241,000	9% (\$110,000)
Carlton Hills-Clinch River 345 KV	Reconductor line	\$5,483,000	4% (\$225,000)
	Total allocated to project		\$9,824,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.